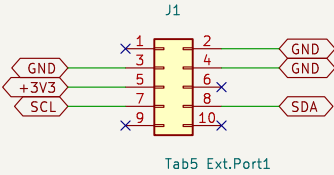
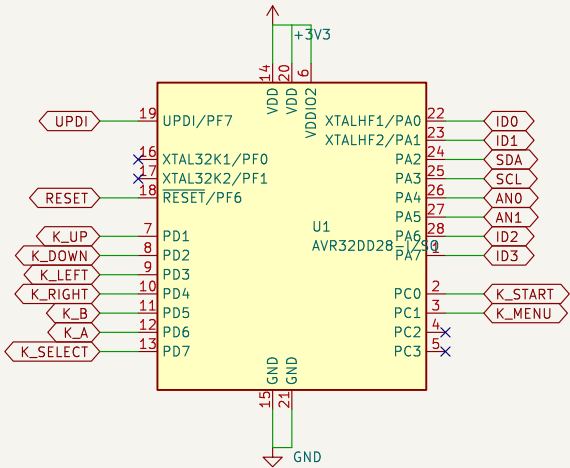


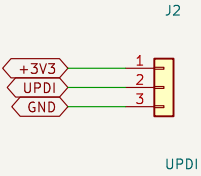
MCU – AVR32DD28 (SOIC–28) running firmware_avrdd (I2C client 0x6D, RetroPad Ext.Port1) – verify pin 1 orientation before fab

Buttons – one MCU pin each, switch to GND (internal pull–up); slot table in firmware_avrdd/pinmap.h

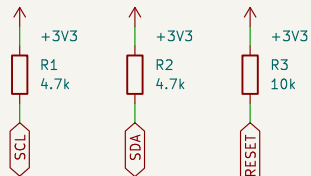
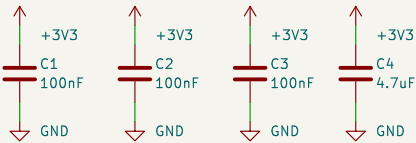
PD1–PD7, PC0–PC3, PF0–PF1: one button per pin (internal pull–ups)



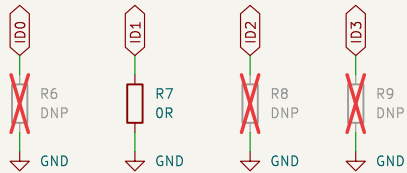
UPDI programming (SerialUPDI / MPLAB Snap)



Decoupling (VDD x2, VDDIO2), I2C pull–ups, RESET pull–up



Console ID straps: 0R to GND = bit set. Game Boy = ID 2 (fit R7; R6, R8, R9 DNP)



Mounting holes (M3, 96 mm apart)

